

# DeFi Arbitrage Scanner (simulation only)

Polls Uniswap V2 and SushiSwap WETH/USDC reserves, computes the profit-maximizing round-trip size, subtracts estimated gas, and logs whether the gap was actually profitable. No transactions, no private keys.

## Run

```
pip install -r requirements.txt
export RPC_URL="https://your-ethereum-rpc"
python scanner.py --polls 5000 --interval 12
```

## Method

- **Constant-product pricing:**  $out = in * 997 * r_{out} / (r_{in} * 1000 + in * 997)$
- Trade size found by ternary search (profit is concave in size)
- Gas = assumed gas units x live gas price x ETH price
- Verify pair addresses on Etherscan before trusting output

## Results (fill in from YOUR run)

- Period / blocks sampled:
- Blocks with gross gap > 0:
- Blocks with net profit > 0 after gas:
- Best / median net opportunity:
- Why most gaps were not profitable:

## Limitations

- Ignores MEV competition, so real capture would be far lower
- Gas units are a rough estimate

- Only two venues and one pair
- Reserves read at block time, not simulated inclusion

## **Ideas to extend**

- Add more pairs and venues (Uniswap V3 ticks)
- Compare against a fork-based simulation (Foundry/Anvil)
- Model private-orderflow bidding to validators